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Simulating Indoor Evacuation of Pedestrians: The Sensitivity of Predictions to Directional-Choice Calibration Parameters

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Abstract

The increasing occurrence of safety-related incidents like fire and terror attacks in crowded public facilities and mass gatherings has heightened the importance of planning for efficient evacuations through optimizing evacuation routes and architectural designs. This calls for the development of simulation and analytical tools that can replicate occupants' responses and thereby their most likely movement patterns. Such models must be accurate to prevent inappropriate design and planning. One major factor connected to prediction accuracy is the sensitivity of modeling outputs to the value of their various parameters. We report on implementation of a calibrated model of directional choices in a microscopic simulation model of pedestrians' evacuation. We show how estimates of the aggregate measures of prediction are sensitive to the parameters of this tactical level (i.e., directional choice) model. Results demonstrate that the prediction of the total evacuation time and average individual evacuation times are closely correlated with one another in terms of their variation, and are both very sensitive to the specification of each directional choice parameter. Simulated evacuation time could vary up to nearly 30% depending on parameter values. The observed sensitivity highlighted the significance of importing well-calibrated parameters into such simulation models and practicing consistent degrees of accuracy for all levels of decision modeling. We also inferred that the two aggregate measures (i.e., total evacuation time and average individual evacuation times) can be used interchangeably as the basis for evacuation optimization or sensitivity analysis practices.

The design of efficient evacuation plans and accessing accurate estimates of evacuation times have become major concerns related to the safety of crowded public facilities, high-rise buildings, and mass gatherings (*1*, *2*). Attention to this problem has heightened as a result of the increasing occurrence of emergency incidents in such environments within recent years (*3*). Research on this problem has grown at a fast pace aiming at developing simulation and optimization tools for supporting fast discharge of occupants (*4*, *5*). Such simulation models are meant to allow planners to evaluate the efficiency of their evacuation plans and the architectural design of facilities, perform reliable risk analysis, and identify problematic locations and critical crowding levels.

The accuracy of such models has always been a major issue given the potential ramifications of inappropriate design or poor planning in this context. Clearly, the accuracy of such models relies heavily on how accurately they replicate the element of human behavior. This has particularly been a significant modeling aspect for the microscopic simulation models that attempt to replicate evacuation patterns by simulating individuals' behavior (as happens in reality) (*6*).

The paucity of data and observations in this particular context has been regarded as a major obstacle towards calibrating and validating many of the existing methodologies and heuristic approaches (*7*, *8*). To address this challenge, one has to resort either to the available field observations from previous emergency incidents (e.g., from CCTV footage) or to collect data in experimental settings. Both approaches, however, pose their own challenges.

While observations from naturally-occurring emergencies are valued for their realism (*9*, *10*), they are rarely

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accessible to researchers or do not provide sufficient material for modeling or model calibration. Such observations are often one-off and do not offer replications, and the analyst does not have any influence on the observations and thus cannot manipulate any factor for a systematic analysis of a specific phenomenon. They also may be subject to high levels of measurement error (e.g., due to low quality or limited coverage of recorded images).

On the other hand, experimental data provides greater flexibility in that the analyst is in charge of all control factors and can vary them through efficient designs and observe the behavior systematically (*11–15*). Measurement errors can be kept to a minimum level, and the analyst can replicate the experiments multiple times and draw robust inferences. The major issue is how realistically the context of an experiment can represent a case of an emergency (*16*). There are certainly limits to how realistic an experiment of evacuation behavior could be, given the ethical and safety restrictions.

A number of authors in this field have suggested that the aspects of the human behavior that need to be investigated and modeled could be classified into three levels of decision making (*17–20*): (a) occupants' reaction times and pre-evacuation activities, known as "strategic" decisions, (b) their directional route choices, known as "tactical" decisions, and (c) their momentary step choices to avoid collision, known as "operational" decisions. It is clear that accurate modeling in this context necessitates consistent modeling accuracy at all three levels.

In a recent review of the empirical studies in this field, Haghani and Sarvi (*21*) quantitatively showed that studies related to operational (also known as "walking behavior") models greatly outnumber studies that have addressed the two other aspects of evacuation modeling. Well-functioning and data-supported walking models have been proposed that can adequately replicate the step choices of evacuees (given their directional preferences) (*22–24*). How occupants make their directional choices has been much less understood, however (*25–27*). A number of studies have resorted to pure theoretical or modeler-defined assumptions for this aspect (*28–31*). In the absence of empirical observations, however, one may question the accuracy of such assumptions which could often be quite arbitrary. This could be partly attributed to the greater challenges involved in experimenting with the strategic and tactical levels of decision making as opposed to observing and analyzing pedestrians' walking.

In a recent literature survey, Haghani and Sarvi (*21*) reviewed the possible relevant data collection methods in this context and identified seven major empirical methods in their categorization: (1) animal crowd experiments (*32*, *33*), (2) virtual reality experiments, (3) controlled laboratory experiments with human crowds, (4) evacuation drills in real buildings, (5) video analysis of naturally-occurring emergencies, (6) video analysis of natural walking, and (7) interview surveys with survivors of mass emergencies. They discussed the advantages and disadvantages offered by each class of data collection and also determined that each class only provides data for particular levels of escape decision making that we categorized earlier.

For the particular problem of directional choices, laboratory experiments with human crowds (*34*), virtual reality experiments (*27*, *35*), and evacuation drills (*36*) have proven most popular in former studies. Experiments with human crowds in artificial environments, in particular, have been regarded as a reasonable compromise between the level of contextual realism and the level of controllability. Despite the great efforts that have been made in recent studies, a missing element has been experimental design and analysis at the level of individuals, as previous studies have predominantly been based on the analysis of aggregate (or macroscopic) measures of behavior (*26*, *37*, *38*). Although useful insight can still be inferred from such aggregate measures, this common feature of the past studies has partly hindered the derivation of behavioral models suitable for microscopic simulation models.

In this study, we report on a series of controlled laboratory experiments with human crowds dedicated to wayfinding and directional choice behavior. The experimental observations have been extracted at disaggregate (i.e., individual) level from which we derived a multivariable model of directional choices. The model probabilistically replicates the directional choices by quantifying evacuees' tradeoffs between a number of factors related to the physical architecture of the escape environment (i.e., physical factors) and a number of factors related to peer influence (i.e., social factors). This tactical level model is then integrated with a conventional operational level model of walking to form a bi-layer simulation model of evacuation. We report on sensitivity analyses on the directional choice parameters of this model to examine the sensitivity of system-wise aggregate measures (e.g., total evacuation times) to these calibration parameters.

Methods

Operational Model Specification

The simulation model of pedestrian evacuations employed in this analysis consists of two layers of decision making. At the operational level, the movement of pedestrians is simulated based on the notion of social force model originally proposed by Helbing, Farkas, and Vicsek (*39*). At the tactical level, the directional route choices of pedestrians are simulated using a probabilistic

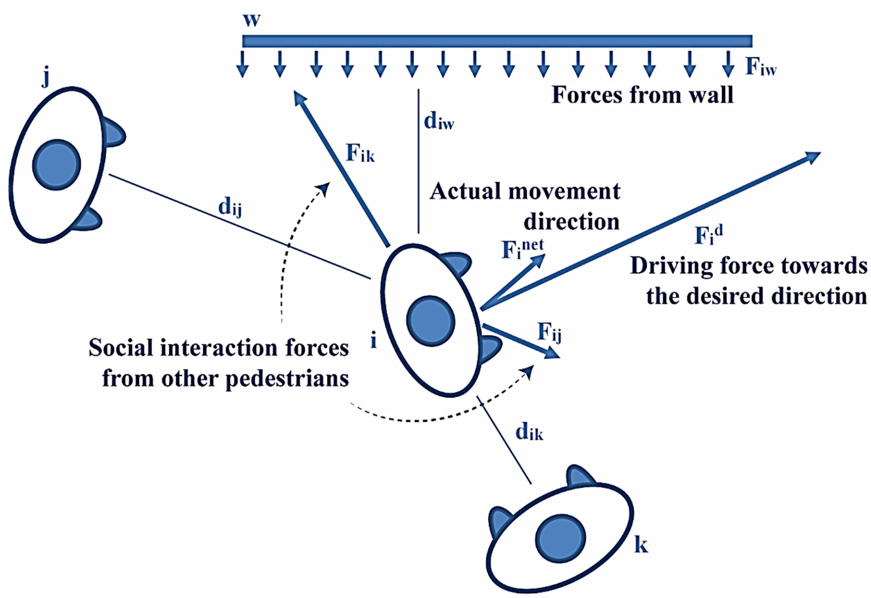

Figure 1. Conceptual illustration of the social force based model for pedestrians' walking, the operational layer of the simulation model employed in this study.

econometric model of discrete choice. The outcome generated from the tactical level model is used as input by the lower level (i.e., the operational level) to generate the simulated movements.

At the operational level model, for a particular pedestrian i, each step motion is determined by the net outcome ($\vec{F}_i^{net}$) of three types of virtual forces: (1) the desired force of pedestrian i driving him/her towards his/her preferred direction $\vec{F}_i^d$, (2) the social interaction forces from other pedestrians $j \neq i$, $\vec{F}_{ij}$; and (3) the forces from walls (w) (or any architectural barrier that the pedestrian would avoid collision with) $\vec{F}_{iw}$ (Equation 1). These three components are defined by Equations 2, 3, and 4 respectively. In these equations, m_i is the body mass of individual i, $\vec{v}_i$ is the vector of velocity for individual i, $\vec{v}_i = d\vec{r}_i/dt$, with $d(.)$ denoting the derivation operator and $\vec{r}_i$ signifying the vector of position for individual i. Also, t represents time and dt signifies the time increment. The unit vector $\vec{e}_i^d$ denotes the desired direction of pedestrian i and v_i^d is the desired (magnitude of) velocity that pedestrian i aims to achieve. The body radius of pedestrian i is denoted by r_i, and d_{ij} denotes the inter-individual Euclidian distance between pedestrians i and j; and e signifies the (natural) exponential function. The function $G\left(r_i + r_j - d_{ij}\right)$ equals $r_i + r_j - d_{ij}$ when $r_i + r_j - d_{ij} > 0$ and equals 0 otherwise. Also $\vec{n}_{ij}$ is defined as $\vec{n}_{ij} = \frac{1}{d_{ij}}\left(\overrightarrow{r_i - r_j}\right) = \left(n_{ij}^1, n_{ij}^2\right)$ indicating the unit vector that connects pedestrian j to i. The term Δv_{ji}^t is defined as $\Delta v_{ji}^t = \left(\vec{v}_j - \vec{v}_i\right).\vec{t}_{ij}$ in which $\vec{t}_{ij} = \left(-n_{ij}^2, n_{ij}^1\right)$ represents the unit vector for tangential direction. Similarly, d_{iw} signifies the distance of pedestrian i to wall w, and $\vec{n}_{iw}$ denotes the unit vector on the direction perpendicular to the wall w from the position of pedestrian i, and $\vec{t}_{iw}$ is the unit vector on the direction perpendicular to $\vec{n}_{iw}$ (similar to the relation between $\vec{n}_{ij}$ and $\vec{t}_{ij}$). Also, τ_i, A_i, B_i, k_1 and k_2 are all calibration parameters. We specified these values according to the article that originally proposed the social force method (*39*). Figure 1 visually conceptualizes the force based operational layer of the model as described in the previous lines.

$$\vec{F}_i^{net} = m_i \frac{d\vec{v}_i}{dt} = \vec{F}_i^d + \vec{F}_{ij} + \vec{F}_{iw} \tag{1}$$

$$\vec{F}_i^d = m_i \frac{v_i^d \vec{e}_i^d - \vec{v}_i}{\tau_i} \tag{2}$$

$$\vec{F}_{ij} = \left(A_i e^{\left(r_i + r_j - d_{ij}\right)/B_i} + k_1 G\left(r_i + r_j - d_{ij}\right)\right) \vec{n}_{ij} + k_2 G\left(r_i + r_j - d_{ij}\right) \Delta v_{ji}^t \vec{t}_{ij} \tag{3}$$

$$\vec{F}_{iw} = \left(A_i e^{\left(r_i - d_{iw}\right)/B_i} + k_1 G\left(r_i - d_{iw}\right)\right) \vec{n}_{iw} - k_2 G\left(r_i - d_{iw}\right)\left(\vec{v}_i \vec{t}_{iw}\right)\vec{t}_{iw} \tag{4}$$

Tactical Model Specification

The operational level layer of the model obtains the component of the desired direction $\vec{e}_i^d$ for each pedestrian i, from the tactical layer (as given input). The tactical model assumes that each pedestrian i perceives a certain level of desirability (or utility) from each exit direction n, U_{ni}, when given a finite and discrete set of N possible exit directions. This utility is composed of a systematic linear-in-parameters component, V_{ni}, and a random error term component, ε_{ni} (Equation 5) added together. In our specification, the systematic part consists of a number of factors related to pedestrian's social interaction as well as a number of factors related to the physical structure of the environment (Equation 6). This term accommodates the pedestrian's tradeoffs between these set of factors. In Equation 6, β_k denotes the utility coefficient (or the relative weight) associated with the kth attribute and x_{nik} is the kth attribute of the alternative exit direction n perceived by (or measured for) pedestrian i, and K is the number of attributes. The error term is assumed to follow an extreme value type I distribution and this distribution is assumed to be identical and independent over all possible exit directions and all simulated pedestrians. Equation 7 determines the probability density function, $f(\varepsilon_{ni})$, of this distribution. This term accommodates measurement errors, perception errors, and model specification errors (e.g., reflecting any variable that is not captured by our systematic part of the model). Based on the premise that the pedestrian would choose the direction with greatest perceived utility, the probability of i choosing any alternative exit direction n will be given by Equation 8.

$$U_{ni} = V_{ni} + \varepsilon_{ni} \quad n = 1, 2, \ldots, N \tag{5}$$

$$V_{ni} = \sum_{k=1}^{K} \beta_k x_{nik} \tag{6}$$

$$f(\varepsilon_{ni}) = e^{-\varepsilon_{ni}} e^{-e^{-\varepsilon_{ni}}} \tag{7}$$

$$P_{ni} = \frac{e^{V_{ni}}}{\sum_{m=1}^{N} e^{V_{mi}}} \tag{8}$$

The systematic part of the utility is specified in our model as Equation 9. The main attributes that are recorded include: $(DIST)_{ni}$, the Euclidian distance of the individual n to exit i measured in meters, $(CONG)_{ni}$ the congestion of exit n for individual i measured in terms of the number of pedestrians (who are present at the vicinity of the exit n, defined as a pre-specified region near the exit), $(FLTOEX)_{ni}$ the size of flow observed by individual i moving towards exit n, measured in terms of the number of pedestrians that are closer to exit n than individual i, are moving towards exit n and are not yet in the vicinity of exit n (i.e., are not counted as congestion), $(VIS)_{ni}$ the visibility status of exit n, for individual i that equals 1 if there is no obstacle or architectural barrier between individual i and the center point of exit n and equals 0 otherwise.

By multiplying $(FLTOEX)_{ni}$ by $(VIS)_{ni}$ and $(1 - VIS)_{ni}$, we basically differentiate between the coefficients of the flows that are moving to visible exits (denoted as $(FLTOVIS)_{ni}$) from the flows moving to the invisible exits (denoted as $(FLTOINVIS)_{ni}$) according to individual i. When exit n is visible to individual i, then $(FLTOVIS)_{ni}$ equals $(FLTOEX)_{ni}$ and $(FLTOINVIS)_{ni}$ becomes zero. Similarly, when exit n is invisible to individual i, then $(FLTOINVIS)_{ni}$ equals $(FLTOEX)_{ni}$ and $(FLTOVIS)_{ni}$ becomes zero. So for each alternative exit n available to each individual i, only one of $(FLTOVIS)_{ni}$ and $(FLTOINVIS)_{ni}$ takes a non-zero value (and that value simply equals $(FLTOEX)_{ni}$). Figure 2 conceptualizes in a visual way the measurements of the social and physical factors related to the tactical layer of the model described above.

$$\begin{aligned} V_{ni} = {} & \beta_1 (DIST)_{ni} + \beta_2 (CONG)_{ni} \\ & + \beta_3 \left(\underbrace{(VIS)_{ni} \times (FLTOEX)_{ni}}_{= (FLTOVIS)_{ni}} \right) \\ & + \beta_4 \left(\underbrace{(1 - VIS)_{ni} {*} (FLTOEX)_{ni}}_{= (FLTOINVIS)_{ni}} \right) \\ & + \beta_5 (VIS)_{ni} \end{aligned} \tag{9}$$

For each pedestrian i, once the probabilities of all available exit directions are generated based on the tactical model, the choice of the exit direction is simulated and the outcome, $\vec{e}_i^{\,d}$, is then introduced to the operational level of the model. Currently, in the analysis we reported in this study, the exit direction choice is a one-off decision and is simulated as soon as the pedestrian steps in an environment (i.e., room). The directional choice parameters are also assumed to be generic and also constant over the population of the simulated pedestrians for this study.

Tactical Model Calibration

The parameters of the tactical layer model described in the previous section were estimated using a dataset of individual directional choice observations collected from a set of (field-type) controlled laboratory experiments. Nearly 150 individuals were recruited and invited to participate in the experiments which took place at a basketball court in Melbourne, Australia. We made a number of temporary built environments using panels designed for the experiments and asked the participants to escape from that environment as if they were escaping from an imminent danger. They were allowed to run and were asked to compete with other participants and choose the best escape direction (using the set of available alternatives) as they would do in an actual case of emergency.

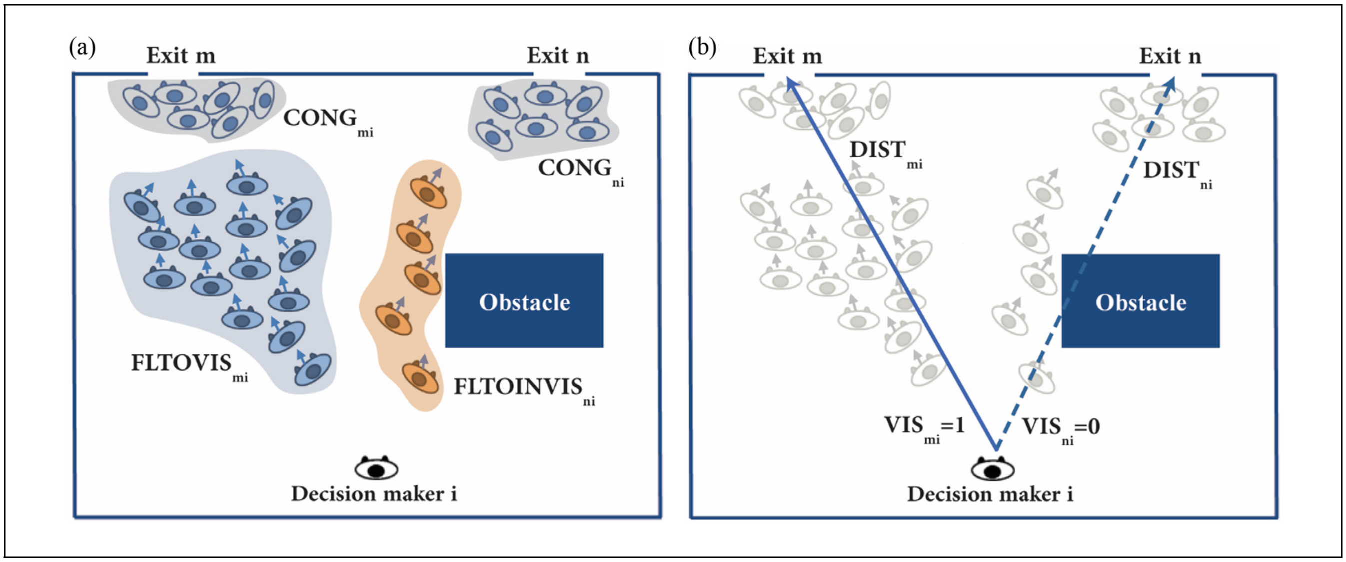

Figure 2. Conceptual illustration of the variable measurements for the directional discrete exit-choice model (the tactical layer model) in the simulation method employed in this study: (*a*) visualization of the measurement of social interaction factors, (*b*) visualization of the measurement of physical factors.

The experiment in total was composed of 29 runs of simulated escape as described above, over which we varied the levels of certain control factors rather randomly in order to create maximum variation in the individual-level observations (*40*). These control factors include the number of evacuees we let in for each scenario, the number and widths of exit doors, and the positioning of available exits. The participants were not informed of these factors (except for the number of evacuees, which they noticed themselves) prior to each run in order to mitigate the effect of habituation to a fixed architectural setting. Figure 3 depicts the blueprint of the environment that was built and manipulated during the experimental scenarios. Figure 4*a* and *b* presents two still images from two scenarios (one with 150 and one with 75 evacuees).

The experiments were videorecorded using a GoPro camera mounted 8 meters above the floor. The legs of the camera tripod were positioned inside the obstacle in the room. The footage was analyzed using PeTrack software to extract the trajectories and the choices of participants. An individual-based data extraction approach was performed in order to provide a modeling dataset from which the parameters of the described tactical model could be estimated. The behavior of every single participant in these scenarios was examined individually, and their trajectories, their choices, and the characteristics of their exit directional choice were extracted. A total of 3,338 observations were extracted. For each observation we recorded three essential elements of the directional choice: (1) the choice set of available exit directions (treated as unlabeled alternatives), (2) the attributes of each exit direction (including the social and physical

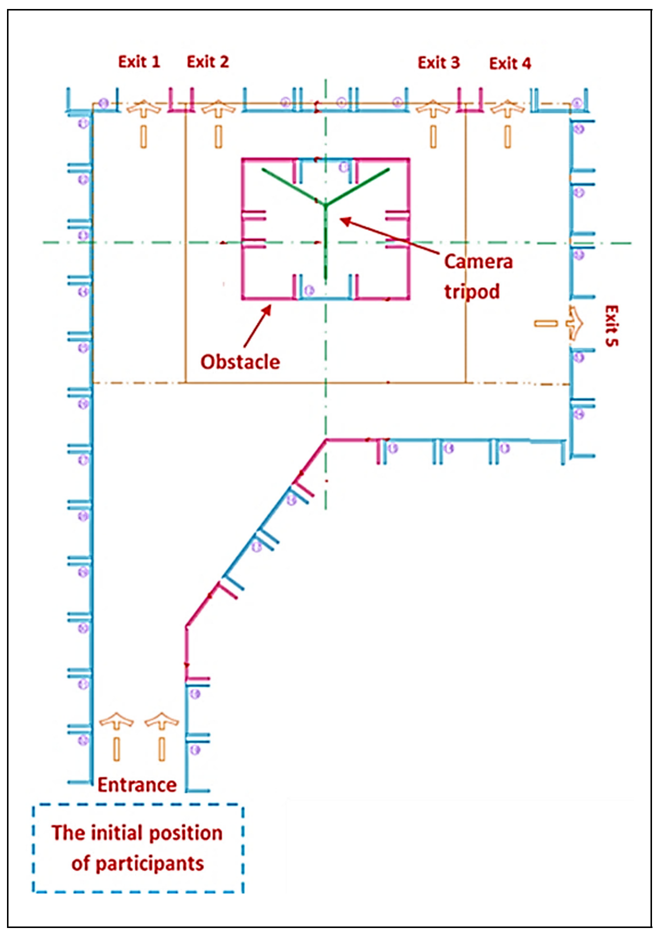

Figure 3. Blueprint of the experimental setup. In each scenario, only a subset of exits was available.

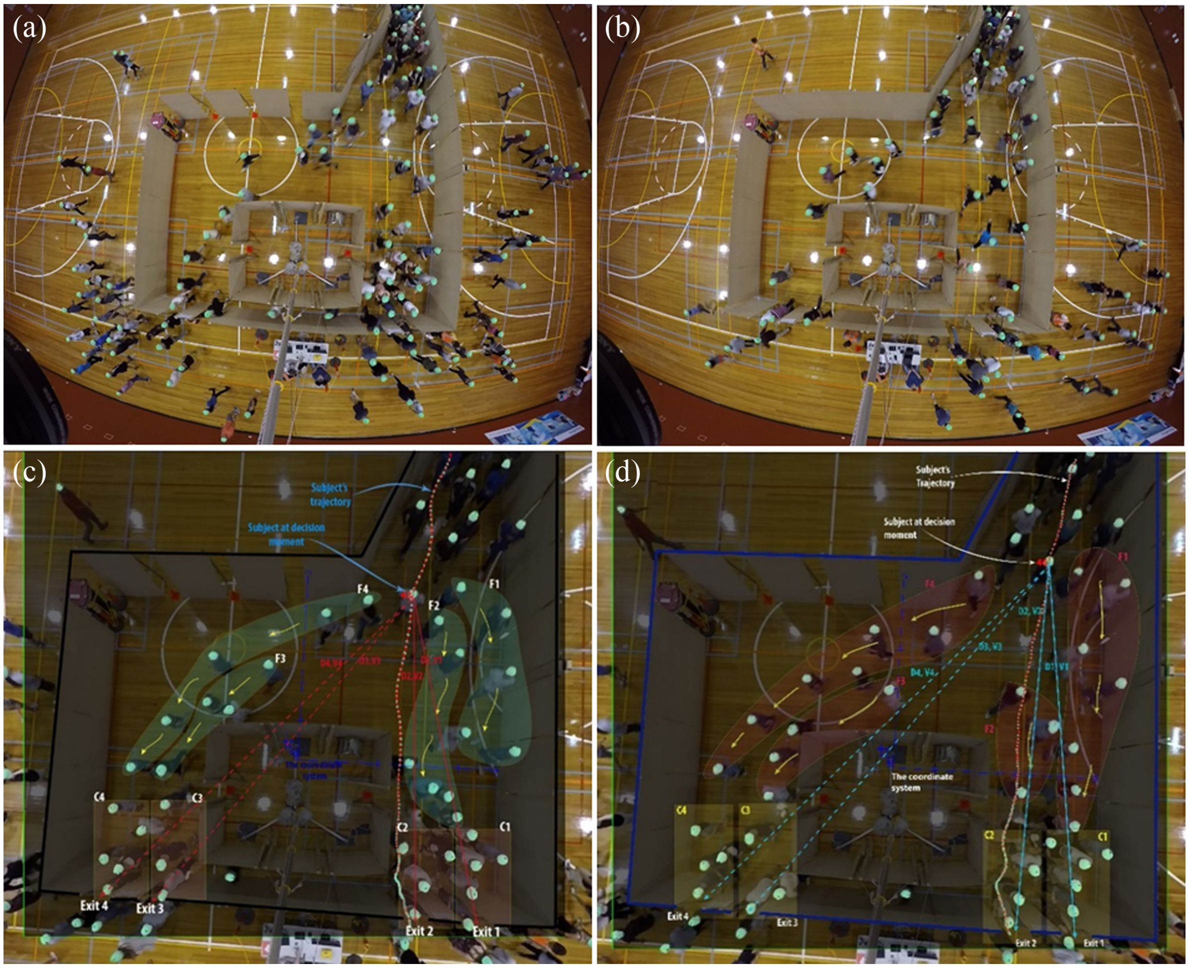

Figure 4. Still images from footage of the simulated evacuation experiments. (*a*) and (*b*) show snapshots from the raw footage of scenarios with 150 and 75 evacuees respectively. (*c*) and (*d*) illustrate the data extraction process at the level of individuals. The process of recording of choice observations is demonstrated by superimposing the visualization of the choice components (consistent with Figure 2) on the interface of the image-processing software. The trajectories show the "choice." The "choice set" and the "alternative attributes" (including the social and physical factors) are illustrated on the pictures. The collection of these three elements qualifies as one disaggregate choice observation.

factors described earlier), and (3) the choice (as the dependent variable). Figure 4*c* and *d* visually demonstrate how these observations were recorded by exemplifying two pedestrians and the characteristics of their choices.

Analyses and Results

Based on the dataset described in the previous section, the mathematical formulation presented in Equation 9 was calibrated using a maximum likelihood estimation process. The calibrated model is presented in Equation 10 where the numbers in subscripts of the estimates represent the *t*-statistic values associated with those estimates. All estimates were statistically significant at conventional levels.

$$V_{ni} = -0.256_{(-11.40)}(DIST)_{ni} - 0.138_{(-12.90)}(CONG)_{ni} - 0.024_{(-2.07)}\left(\underbrace{(VIS)_{ni}*(FLTOEX)_{ni}}_{=(FLTOVIS)_{ni}}\right) + 0.093_{(9.52)}\left(\underbrace{(1-VIS)_{ni}*(FLTOEX)_{ni}}_{=(FLTOINVIS)_{ni}}\right) + 0.710_{(2.85)}(VIS)_{ni} \quad (10)$$

The combined microscopic simulation model was then investigated by performing a sensitivity analysis on the values of these parameters. The architectural setup presented in Figure 5 was used as the basis of this analysis.

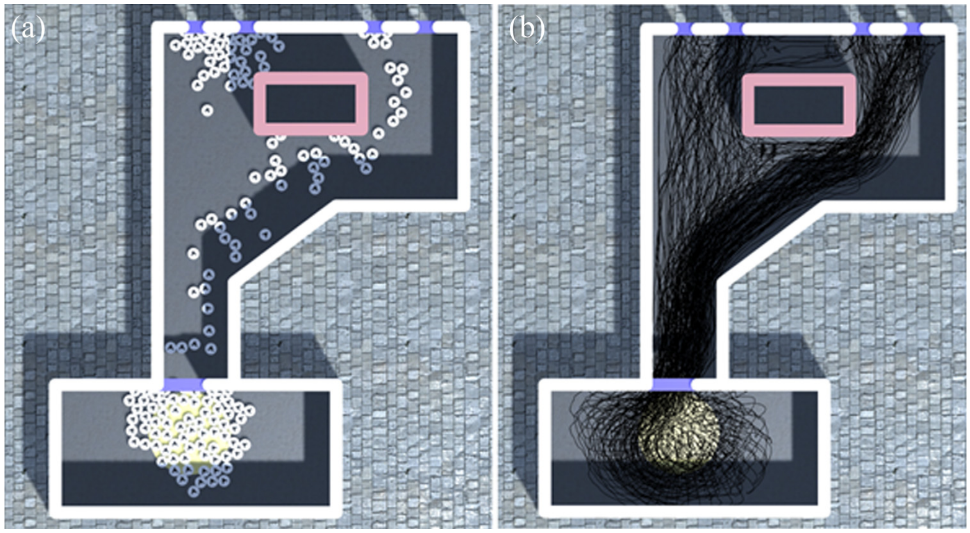

Figure 5. Still images from the visualization of the simulation model employed in this study and the architectural setup used for the sensitivity analyses. (*a*) simulation process of 150 pedestrians while calculations are in progress, (*b*) trajectories of all simulated pedestrians once a simulation run is over.

Figure 5 illustrates still images from the visualization of a simulated scenario while the simulation calculation is in progress as well as an image depicting the trajectories of all simulated pedestrians at the end of the simulation calculations for that scenario. For the sensitivity analysis, 150 simulated pedestrians were generated all at the same time in the rectangular area shown in the pictures. Two aggregate measures of performance were recorded for each run of the simulation: (1) the total evacuation time and (2) the average of individuals' evacuation times. Sensitivity analyses were performed by taking each parameter of the tactical model and departing from its estimated value in both increasing and decreasing directions to the level that no further sensitivity was observed. Given the fact that the variables of the tactical model have different units of measurement, the ranges of sensitivity for different parameters were different. Therefore, no fixed common range for changing the level of parameters could be set a priori.

The simulated experiments associated with the sensitivity analysis of each parameter were performed while the values of all other parameters were kept constant at their estimated values. For the parameter under investigation, the values were gradually increased and decreased while recording the two aforesaid measures. For each individual parameter value, 50 repetitions of the simulation were performed. A total of nearly 7,500 simulation repetitions was practiced for these analyses.

The results of these analyses are summarized in Figures 6–10, where the values of "total evacuation times" (left vertical axis) and "average individual evacuation times" (right vertical axis) have been averaged over 50 simulation repetitions and have been visualized for each parameter value as line charts. The error envelopes represent the standard deviation of the measurements. The graphs are respectively related to the analysis on the values of the parameters β_1 to β_5 in Equation 9. Also, each graph visualizes the estimated value for that parameter along with the 95% confidence interval of that estimate as vertical lines superimposed on each plot.

A common observation based on these sensitivity analyses reflected in Figures 6–10 is the consistency between the two aggregate measures of performance that we applied, total evacuation time, and average of individual evacuation times. As can be seen in these plots, these two measures always invariably display the same type of "sensitivity patterns" to the values of the behavioral parameters. The choice of the most relevant aggregate measure of performance for evacuations is important when system-wise optimization (of architectural design or evacuation routes) becomes the analyst's concern. This high level of correlation between these two measures of performance could be used as evidence that either of these measures could be alternatively applied as the objective function of such optimization analyses.

Another common observation in our analyses was that, the outcome of the simulation is only sensitive to the values of each of the directional parameters within a certain range that varies from parameter to parameter. Also, the values of the estimates for these parameters fell invariably within this sensitive range for all five attributes. In our experience, for some attributes, particularly for the physical attributes whose values are not as time dependent as the social attributes and do not change much by

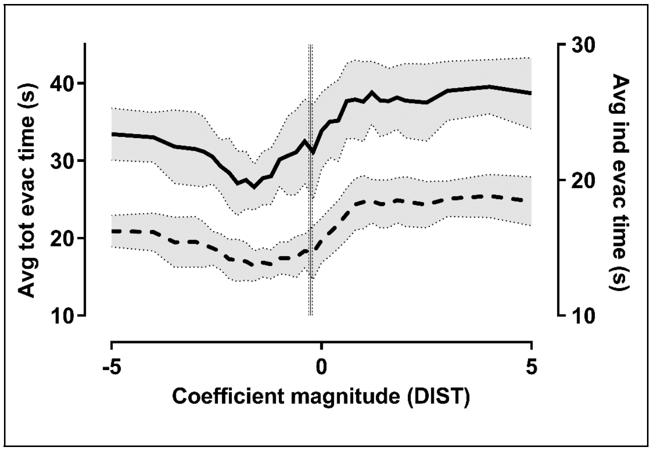

Figure 6. The variation of total evacuation times (solid line, represented by left vertical axis) and average individual evacuation times (dashed line, represented by right vertical axis) in response to the change of the parameter for DIST in the tactical level of the model. Each point on the charts has been obtained by averaging 50 simulation repetitions. The error bands represent the standard deviation of the simulated observations. The vertical line shows the calibrated value of the parameter and the dashed vertical lines specify the 95% confidence interval for that estimate.

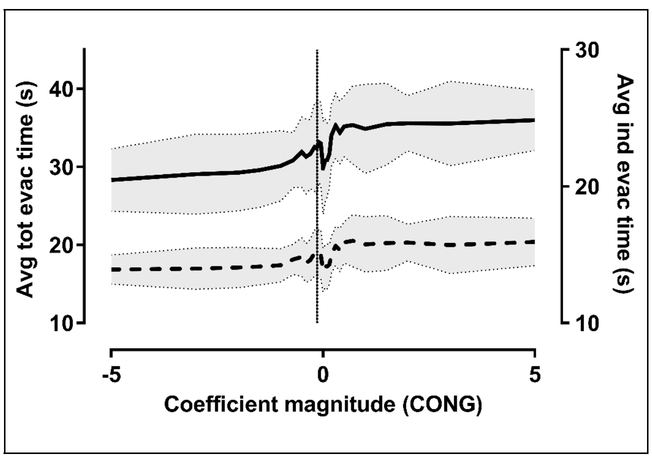

Figure 7. The variation of total evacuation times (solid line, represented by left vertical axis) and average individual evacuation times (dashed line, represented by right vertical axis) in response to the change of the parameter for CONG in the tactical level of the model. Each point on the charts has been obtained by averaging 50 simulation repetitions. The error bands represent the standard deviation of the simulated observations. The vertical line shows the calibrated value of the parameter and the dashed vertical lines specify the 95% confidence interval for that estimate.

the dynamics of the system (take VIS for example), once the value of the parameter associated with an attribute is taken to an "extreme" level for that attribute, that attribute becomes dominant. Once this happens, a very similar pattern of movement is observed at each repetition, as if the directional choices are virtually deterministic.

We observed in our simulated experiments that the level of sensitivity of the aggregate performance measures to the parameter values varies from attribute to attribute. For instance, depending on how we specify the value of DIST parameter, the prediction of total evacuation time for our simulated setup can vary from nearly 27 s to nearly 40 s. Among the attributes included in our tactical model, the least degree of sensitivity of evacuation performance was observed for CONG. Also, it is observable that, except for CONG, the pattern of the sensitivity of the evacuation time to the parameter values is not monotonic and they each show a minimum at an intermediate level for that parameter.

Another aspect of this analyses is connected to the fact that each of the parameters investigated here conveys behavioral interpretation and is reflective of how a "typical" evacuee prioritizes these factors when choosing the direction of movement. Similarly the values of our parameter estimates inferred from our experimental observations are reflective of the dominant prioritization that subjects displayed in the experimental setting. The question one may raise is whether the observed "natural" prioritization of evacuees was optimal from a system optimization perspective. What if the evacuees had shown a different type of prioritization (or relative decision weights)? Could that have benefited the system in terms of the total evacuation times? Also, another relevant question is whether a multi-attribute tradeoff is more beneficial than a single-attribute decision. In other words, would it be beneficial from a system perspective if all evacuees make their decisions based on choosing the closest exit direction, or make their decisions based on only choosing among the visible exit directions, or make their decisions based on following the majority?

The answers to the above questions are case specific, as shown in Figures 6–10. According to Figure 7, avoiding congestion at exits is always beneficial from the system perspective. Decreasing the value of this parameter (equivalent of reinforcing a congestion-avoidance tendency) resulted almost consistently in shorter evacuation times. For the other four parameters, however, the minimum evacuation time was never achieved at extreme values. According to Figure 6, making decisions dominantly based on minimizing distance (associated with an extremely negative parameter value for DIST) would benefit the system more than making decisions dominantly based on choosing the furthest exit directions (associated with an extremely positive parameter value for DIST). The minimum evacuation time, however, occurred at neither of these extreme values, but rather at an intermediate level. According to Figure 8, for visible exits, complete avoidance or complete imitation of the behavior of moving peers choosing that direction are almost equally non-beneficial, and the optimum is again

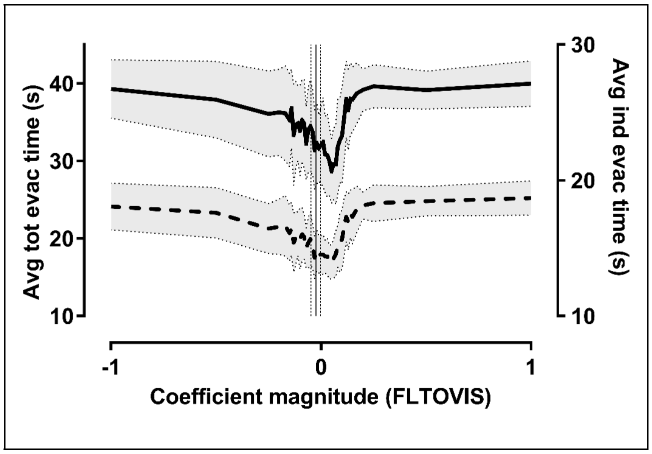

Figure 8. The variation of total evacuation times (solid line, represented by left vertical axis) and average individual evacuation times (dashed line, represented by right vertical axis) in response to the change of the parameter for FLTOVIS in the tactical level of the model. Each point on the charts has been obtained by averaging 50 simulation repetitions. The error bands represent the standard deviation of the simulated observations. The vertical line shows the calibrated value of the parameter and the dashed vertical lines specify the 95% confidence interval for that estimate.

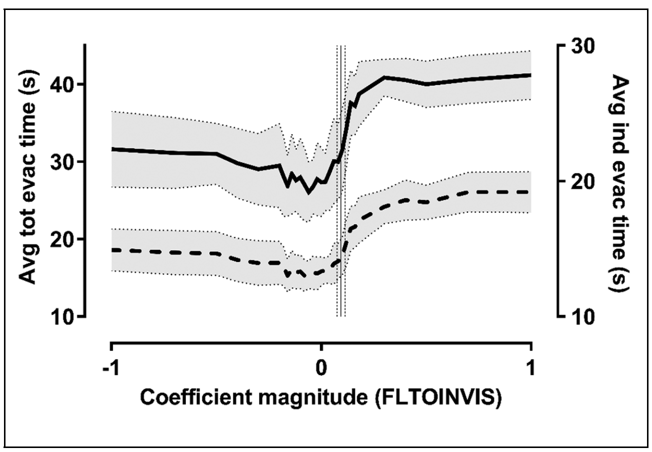

Figure 9. The variation of total evacuation times (solid line, represented by left vertical axis) and average individual evacuation times (dashed line, represented by right vertical axis) in response to the change of the parameter for FLTOINVIS in the tactical level of the model. Each point on the charts has been obtained by averaging 50 simulation repetitions. The error bands represent the standard deviation of the simulated observations. The vertical line shows the calibrated value of the parameter and the dashed vertical lines specify the 95% confidence interval for that estimate.

at an intermediate level. Whereas, according to Figure 9, for invisible exit directions, complete avoidance (extreme negative value) is less suboptimum than complete imitation of the moving peer evacuees choosing that direction. According to Figure 10, making directional decisions purely based on choosing between visible exits (extreme

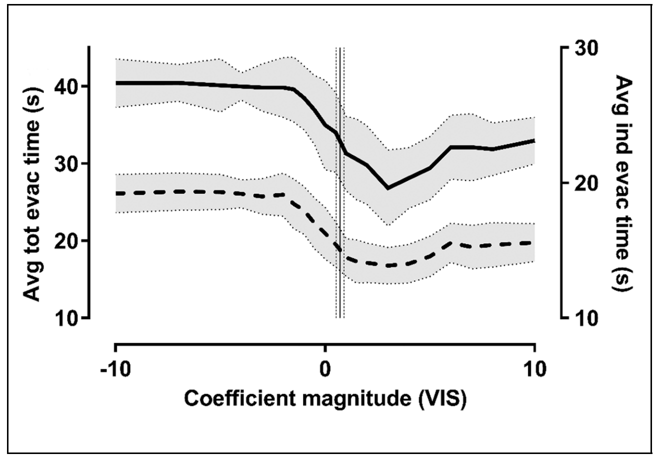

Figure 10. The variation of total evacuation times (solid line, represented by left vertical axis) and average individual evacuation times (dashed line, represented by right vertical axis) in response to the change of the parameter for VIS in the tactical level of the model. Each point on the charts has been obtained by averaging 50 simulation repetitions. The error bands represent the standard deviation of the simulated observations. The vertical line shows the calibrated value of the parameter and the dashed vertical lines specify the 95% confidence interval for that estimate.

positive values) is also less suboptimal than making decisions purely based on choosing between invisible exits and yet again, an optimum tendency is observable at intermediate levels.

It is also clear that, in terms of the behavior (i.e., set of directional parameters) that is mathematically optimum and results in minimum evacuation times in our setup, there may not exist any unique solution and our analysis was not meant to find such a set of parameters either. It was, however, reflected indirectly in our analysis that for each of the behavioral parameters at tactical level, assuming all other parameters fixed at their estimated values, there could be certain behavioral tendencies (i.e., certain set of attribute prioritization) that could result in better system performance (in terms of the aggregate measures of evacuation).

Summary and Conclusions

This paper reports on the calibration of the directional choice parameters of a microscopic simulation model for pedestrian evacuations using a large data set of disaggregate experimental observations. An analysis was performed on the sensitivity of the simulated evacuation time estimates to the value of these parameters.

The results showed a great degree of sensitivity for the evacuation time estimates to the specification of these behavioral parameters, highlighting the importance of accurate calibration for such parameters. The results also showed that the variation of total evacuation time and average individual evacuation time estimates are closely correlated, suggesting they can be almost indifferently used as the objective function for optimization purposes.

From the behavioral perspective, the simulated results indicated that directional choice-making behavior based on single attributes is less beneficial to the system than multi-attribute tradeoffs (which was the type of behavior people actually displayed in our experimental observations). Extreme values for the parameters proved almost invariably suboptimal from a system perspective, and for each parameter, the value corresponding with the best system performance (minimum evacuation time) occurred at an intermediate level. In other words, and as a practical example, the theoretical microscopic models that replicate the directional route choices purely based on the shortest distance criterion (*29*) may actually result in overestimation of the evacuation time. The same argument can be applied to the models that use "follow-the-crowd" behavior (*41*, *42*) as the directional choice criterion.

The results emphasize the importance of accurate calibration of simulation models for evacuation as opposed to the less costly approach of applying intuitive or pure heuristic rules of behavior. The type of sensitivity analysis reported in this work is relatively time consuming given the very large number of simulation repetitions needed to collect the required data. However, there is potential for this type of analysis to lead to the identification of the optimum behavioral tendencies, which could also have important practical applications. As a simple question, one may ask whether evacuation managers should advise people to follow each other in case of an emergency or whether it is more beneficial for the occupants to avoid heavily used paths (the same question could also be raised for choosing the nearest exit). Answering these types of question was not the primary purpose of this study and this aspect is pointed out as a side outcome of this analysis as preliminary insight. We assume that there is no short universal answer to the problem of "optimum behavior" and that it may depend on many factors, such as the architectural and geometric features of the environment as well as the level of crowding in the environment (and possibly other factors). We intend to provide more comprehensive analysis on these aspects in subsequent studies. Finally, as pertinently pointed out by a referee of this paper, it is crucial to have an understanding of the universality of the parameters inferred from the laboratory experiments that we proposed. The extent to which these parameter estimates are the artefact of the physical setup of the experiment is a valid question that is worth investigation through further experimentation. The authors will report on their findings about the parameter transferability in a future work.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: M. Haghani, M. Sarvi; data collection: M. Haghani, M. Sarvi; analysis and interpretation of results: M. Haghani; draft manuscript preparation: M. Haghani, M. Sarvi, A. Rajabifard. All authors reviewed the results and approved the final version of the manuscript.

References

1. Illiyas, F. T., S. K. Mani, A. P. Pradeepkumar, and K. Mohan. Human Stampedes During Religious Festivals: A Comparative Review of Mass Gathering Emergencies in India. *International Journal of Disaster Risk Reduction*, Vol. 5, 2013, pp. 10–18.
2. Yogameena, B., and C. Nagananthini. Computer Vision Based Crowd Disaster Avoidance System: A Survey. *International Journal of Disaster Risk Reduction*, Vol. 22, 2017, pp. 95–129.
3. Kobes, M., I. Helsloot, B. de Vries, and J. G. Post. Building Safety and Human Behaviour in Fire: A Literature Review. *Fire Safety Journal*, Vol. 45, No. 1, 2010, pp. 1–11.
4. Vermuyten, H., J. Beliën, L. De Boeck, G. Reniers, and T. Wauters. A Review of Optimisation Models for Pedestrian Evacuation and Design Problems. *Safety Science*, Vol. 87, 2016, pp. 167–178.
5. Yu, Z., D. Li, S. Zhu, W. Luo, Y. Hu, and L. Yuan. Multisource Multisink Optimal Evacuation Routing with Dynamic Network Changes: A Geometric Algebra Approach. *Mathematical Methods in the Applied Sciences*, Vol. 41, No. 11, 2018, pp. 4179–4194.
6. Duives, D. C., W. Daamen, and S. P. Hoogendoorn. State-of-the-art Crowd Motion Simulation Models. *Transportation Research Part C: Emerging Technologies*, Vol. 37, 2013, pp. 193–209.
7. Shi, L., Q. Xie, X. Cheng, L. Chen, Y. Zhou, and R. Zhang. Developing a Database for Emergency Evacuation Model. *Building and Environment*, Vol. 44, No. 8, 2009, pp. 1724–1729.
8. Haghani, M., and M. Sarvi. Stated and Revealed Exit Choices of Pedestrian Crowd Evacuees. *Transportation Research Part B: Methodological*, Vol. 95, 2017, pp. 238–259.
9. Yang, X., Z. Wu, and Y. Li. Difference Between Real-life Escape Panic and Mimic Exercises in Simulated Situation with Implications to the Statistical Physics Models of Emergency Evacuation: The 2008 Wenchuan Earthquake. *Physica A: Statistical Mechanics and Its Applications*, Vol. 390, No. 12, 2011, pp. 2375–2380.
10. Bernardini, G., E. Quagliarini, and M. D'Orazio. Towards Creating a Combined Database for Earthquake Pedestrians' Evacuation Models. *Safety Science*, Vol. 82, 2016, pp. 77–94.
11. Haghani, M., and M. Sarvi. Social Dynamics in Emergency Evacuations: Disentangling Crowd's Attraction and Repulsion Effects. *Physica A: Statistical Mechanics and Its Applications*, Vol. 475, 2017, pp. 24–34.
12. Bode, N. W. F., S. Holl, W. Mehner, and A. Seyfried. Disentangling the Impact of Social Groups on Response Times

and Movement Dynamics in Evacuations. *PLoS One*, Vol. 10, No. 3, 2015, p. e0121227.
13. Wagoum, A. K., A. Tordeux, and W. Liao. Understanding Human Queuing Behaviour at Exits: an Empirical Study. *Royal Society Open Science*, Vol. 4, No. 1, 2017, p. 160896.
14. Shahhoseini, Z., M. Sarvi, and M. Saberi. Pedestrian Crowd Dynamics in Merging Sections: Revisiting the "Faster-is-slower" Phenomenon. *Physica A: Statistical Mechanics and its Applications*, Vol. 491, 2018, pp. 101–111.
15. Shahhoseini, Z., M. Sarvi, M. Saberi, and M. Haghani. Pedestrian Crowd Dynamics Observed at Merging Sections. *Transportation Research Record: Journal of the Transportation Research Board*, 2017. 2622: 48–57.
16. Haghani, M., and M. Sarvi. Human Exit Choice in Crowded Built Environments: Investigating Underlying Behavioural Differences Between Normal Egress and Emergency Evacuations. *Fire Safety Journal*, Vol. 85, 2016, pp. 1–9.
17. Asano, M., T. Iryo, and M. Kuwahara. Microscopic Pedestrian Simulation Model Combined with a Tactical Model for Route Choice Behaviour. *Transportation Research Part C: Emerging Technologies*, Vol. 18, No. 6, 2010, pp. 842–855.
18. Haghani, M., and M. Sarvi. Pedestrian Crowd Tactical-level Decision Making During Emergency Evacuations. *Journal of Advanced Transportation*, Vol. 50, No. 8, 2016, pp. 1870–1895.
19. Duives, D., and H. Mahmassani. Exit Choice Decisions During Pedestrian Evacuations of Buildings. *Transportation Research Record: Journal of the Transportation Research Board*, 2012. 2316: 84–94.
20. Haghani, M., M. Sarvi, and Z. Shahhoseini. Accommodating Taste Heterogeneity and Desired Substitution Pattern in Exit Choices of Pedestrian Crowd Evacuees Using a Mixed Nested Logit Model. *Journal of Choice Modelling*, Vol. 16, 2015, pp. 58–68.
21. Haghani, M., and M. Sarvi. Crowd Behaviour and Motion: Empirical Methods. *Transportation Research Part B: Methodological*, Vol. 107, 2018, pp. 253–294.
22. Moussaïd, M., D. Helbing, and G. Theraulaz. How Simple Rules Determine Pedestrian Behavior and Crowd Disasters. *Proceedings of the National Academy of Sciences of the United States of America*, Vol. 108, 2011, pp. 6884–6888.
23. Robin, T., G. Antonini, M. Bierlaire, and J. Cruz. Specification, Estimation and Validation of a Pedestrian Walking Behavior Model. *Transportation Research Part B: Methodological*, Vol. 43, No. 1, 2009, pp. 36–56.
24. Seitz, M. J., N. W. F. Bode, and G. Köster. How Cognitive Heuristics Can Explain Social Interactions in Spatial Movement. *Journal of the Royal Society Interface*, Vol. 13, No. 121, 2016, p. 20160439.
25. Haghani, M., and M. Sarvi. Identifying Latent Classes of Pedestrian Crowd Evacuees. *Transportation Research Record: Journal of the Transportation Research Board*, 2016. 2560: 67–74.
26. Bode, N. W. F., and E. A. Codling. Human Exit Route Choice in Virtual Crowd Evacuations. *Animal Behaviour*, Vol. 86, No. 2, 2013, pp. 347–358.
27. Lovreglio, R., A. Fonzone, and L. dell'Olio. A Mixed Logit Model for Predicting Exit Choice During Building Evacuations. *Transportation Research Part A: Policy and Practice*, Vol. 92, 2016, pp. 59–75.
28. Lo, S. M., H. C. Huang, P. Wang, and K. K. Yuen. A Game Theory Based Exit Selection Model for Evacuation. *Fire Safety Journal*, Vol. 41, No. 5, 2006, pp. 364–369.
29. Kemloh Wagoum, A. U., A. Seyfried, and S. Holl. Modeling the Dynamic Route Choice of Pedestrians to Assess the Criticality of Building Evacuation. *Advances in Complex Systems*, Vol. 15, No. 07, 2012, p. 1250029.
30. Fu, L., W. Song, W. Lv, and S. Lo. Simulation of Exit Selection Behavior Using Least Effort Algorithm. *Transportation Research Procedia*, Vol. 2, 2014, pp. 533–540.
31. Abdelghany, A., K. Abdelghany, H. Mahmassani, H. Al-Ahmadi, and W. Alhalabi. Modeling the Evacuation of Large-Scale Crowded Pedestrian Facilities. *Transportation Research Record: Journal of the Transportation Research Board*, 2010. 2198: 152–160.
32. Shahhoseini, Z., and M. Sarvi. Collective Movements of Pedestrians: How We Can Learn from Simple Experiments with Non-human (Ant) Crowds. *PLoS One*, Vol. 12, No. 8, 2017, p. e0182913.
33. Shahhoseini, Z., M. Sarvi, and M. Saberi. Insights Toward Characteristics of Merging Streams of Pedestrian Crowds Based on Experiments with Panicked Ants. *Transportation Research Record: Journal of the Transportation Research Board*, 2016. 2561: 81–88.
34. Haghani, M., and M. Sarvi. Following the Crowd or Avoiding It? Empirical Investigation of Imitative Behaviour in Emergency Escape of Human Crowds. *Animal Behaviour*, Vol. 124, 2017, pp. 47–56.
35. Moussaïd, M., M. Kapadia, T. Thrash, R. W. Sumner, M. Gross, D. Helbing, and C. Hölscher. Crowd Behaviour During High-stress Evacuations in an Immersive Virtual Environment. *Journal of the Royal Society Interface*, Vol. 13, No. 122, 2016, p. 20160414.
36. Kobes, M., I. Helsloot, B. de Vries, J. G. Post, N. Oberijé, and K. Groenewegen. Way Finding During Fire Evacuation; An Analysis of Unannounced Fire Drills in a Hotel at Night. *Building and Environment*, Vol. 45, No. 3, 2010, pp. 537–548.
37. Haghani, M., and M. Sarvi. How Perception of Peer Behaviour Influences Escape Decision Making: The Role of Individual Differences. *Journal of Environmental Psychology*, Vol. 51, 2017, pp. 141–157.
38. Heliövaara, S., J.-M. Kuusinen, T. Rinne, T. Korhonen, and H. Ehtamo. Pedestrian Behavior and Exit Selection in Evacuation of a Corridor – An Experimental Study. *Safety Science*, Vol. 50, No. 2, 2012, pp. 221–227.
39. Helbing, D., I. Farkas, and T. Vicsek. Simulating Dynamical Features of Escape Panic. *Nature*, Vol. 407, No. 6803, 2000, p. 487.
40. Haghani, M., M. Sarvi, Z. Shahhoseini, and M. Boltes. How Simple Hypothetical-Choice Experiments Can Be Utilized To Learn Humans' Navigational Escape Decisions in Emergencies. *PLoS One*, Vol. 11, No. 11, 2016, p. e0166908.

41. Low, D. J. Statistical Physics: Following the Crowd. *Nature*, Vol. 407, No. 6803, 2000, pp. 465–466.
42. Haghani, M., M. Sarvi, and A. Rajabifard. Common Misconceptions about Herd-type Behavior in Emergency Evacuations of Pedestrian Crowds. Presented at 96th Annual Meeting of the Transportation Research Board, Washington, D.C., 2017.

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